

MARCO A. GUZMAN JR.

Chemical Engineer · Process Development & Manufacturing Engineering · Capital Projects · Food Manufacturing
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PROFESSIONAL SUMMARY

Chemical engineer with 14 years across process and manufacturing engineering, capital project delivery, and director-level technical leadership. Currently drives process development and optimization on high-volume filling and packaging lines at Kraft Heinz Winchester, an FDA-regulated food plant, taking problems from definition through structured plant trials, data analysis, capital justification, and handoff to production. Has led a \$3M process relocation, installation, and start-up project end to end and authored a five-year capital and technology plan for a production system. Works cross-functionally with Operations, Quality, Maintenance, Engineering, and controls every day, and presents cost, ROI, and technology recommendations to senior business-unit leadership. M.S. Engineering Management (Penn State, 2026), B.S. Chemical Engineering, Lean Six Sigma Green Belt, FE / EIT. Bilingual English and Spanish.

AREAS OF EXPERTISE

Process Development & Optimization · Evaluation of New Processing Technologies · Plant Trial Design, Test-Run Review & Results Analysis · Process Verification, Centerlining & Handoff to Production · Capital Project Management (Scope → Appropriation → Installation → Start-Up) · Cost Estimating, Budget Management & ROI Justification · Project Tracking, Documentation & Reporting · Root Cause Analysis & FMEA · OEE & Loss-Tree Analysis · Lean, Six Sigma, Kaizen, PDCA · Statistical Process Control (Minitab) · Manufacturing Analytics (Grafana, PostgreSQL, Power BI) · Regulatory Compliance (FDA / GMP food manufacturing, OSHA, EPA, PSM) · Contractor Supervision · Cross-Functional & Multicultural Team Leadership · Technical Training & Executive Communication (English / Spanish)

PROFESSIONAL EXPERIENCE

Kraft Heinz · Winchester, VA

April 2023 – Present

Continuous Improvement / Operational Excellence Team Leader

- Own process development and optimization for System 5 at a 29-filler, five-system FDA-regulated food plant, partnering daily with Operations, Quality, Maintenance, and Engineering to expand processing capability, raise equipment performance, and lower conversion cost across pre-filling, filling, and packaging.
- Designed and executed a six-filler leak-isolation trial on the System 5 chill-bath heat exchangers that confirmed cracked plates on 5 of 6 units, and built a glycol mass-balance and pressure-differential model reconciling a 900 gal/week loss. The results defined the repair scope, de-risked the capital commitment, and restoring that cooling capability became a key driver of the line's output and OEE gains.
- Identified, evaluated, and justified new processing equipment: built and presented the capital appropriation and ROI case for filler-bowl replacement across multiple systems to senior business-unit leadership, covering process and equipment-condition analysis, cost estimates, projected savings, and implementation timing.
- Authored System 5's 2025 to 2029 five-year capital and process technology plan charting the line from 69% to 85%+ OEE, supported by a loss tree quantifying a \$7.11M annual yield-loss baseline and isolating \$3.92M/yr of empty-pouch scrap as the top recovery target.
- Delivered a 25% increase in daily true output from 2023 to 2025, reaching the line's record annual volume of 11.3M units in 13% fewer run days (281 down to 244).
- Lead the Physical Infrastructure kaizen team inside the plant-wide Road to 85% OEE program: co-scoped 10 project buckets across compressed air, chillers, glycol, condensate return, and waste, and run the working group that drives them through Gemba data gathering, OEE-impact analysis, and CAPEX justification on quarterly timelines.
- Led a cross-functional process improvement initiative on manual dosing across 29 fillers: traced the root cause to secondary downtime cascading into bowl-temperature drops, drove PLC tag corrections with controls and historian support to restore data integrity, built filler-level tracking and dashboards used in weekly cross-functional reviews, and presented the capital ROI case to senior leadership.
- Delivered all KHMS Phase 0 and Phase 1 operating standards for System 5, including centerline management on the fillers: defining, validating, and locking optimal operating parameters so process deviations are caught and corrected on the floor before they become downtime or quality losses.

- Led a cross-departmental initiative with Training, Quality, and Sanitation to update standardized work instructions and process documentation, and sustained over 97% Daily Management System compliance after a plant digital rollout through operator coaching and training programs.

The Chlorine Institute · Arlington, VA

November 2015 – August 2022

Director of Product Stewardship and Analysis · promoted from Project Engineer → Senior Project Engineer → Director

- Led product stewardship for the chlorine industry's ~200 member companies, staffing and directing the Product Stewardship Issue Team of industry experts, analysts, and engineers chartered to drive end-user chemical accidents toward zero.
- Directed full-lifecycle development of the CHLOREP® Mobile App for hazardous-material emergency responders as Product Owner, owning scope, schedule, budget, launch, and continuous improvement, and translating EPA and OSHA regulatory requirements into delivered technical standards and product features with external developers and internal stakeholders.
- Evaluated emerging technologies and monitored U.S. EPA regulatory activity affecting chlorinated and disinfecting technologies, advocating sound-science positions, and secured recurring funding by presenting data-driven roadmaps, interim reports, and performance metrics to senior executive boards.
- Directed development of end-user safety resources and technical training seminars contributing to a 15% reduction in industry OSHA recordables, and built the member-company incident databases behind the industry's annual safety performance report, cited by Bloomberg.
- Represented the Institute on the World Chlorine Council's Global Science and Efficacy Team and delivered technical end-user training in Spanish through Clorosur across Latin America.
- Led cross-company expert teams and managed member-company stakeholder relationships without direct reporting authority, presenting technical findings to executive boards and industry audiences.

INFRA, S.A. de C.V. · Mexico City, Mexico

December 2012 – October 2015

Reengineering Supervisor

- Led the \$3M Magdalene Project end to end, from appropriation and financial forecasting through scheduling, field execution, installation, and start-up, relocating critical production assets and 150+ personnel on schedule and within budget.
- Supervised on-site construction contractors and cross-functional employee teams across multiple trades, enforcing strict safety compliance while protecting design integrity and quality standards.
- Engineered and validated scalable plant layouts and equipment specifications in AutoCAD, determining equipment capability and integrating complex manufacturing systems into cohesive facility designs.

EDUCATION

M.S., Engineering Management · The Pennsylvania State University

2026

Capstone: engineering-management analysis of the Boeing 737 MAX program (PEST, SWOT, process FMEA, decision analysis); authored a 1,000-iteration Monte Carlo / EMV model in @RISK on training capital alternatives.

B.S., Chemical Engineering · California State Polytechnic University, Pomona (ABET-accredited)

2012

CERTIFICATIONS & TECHNICAL SKILLS

Certifications: Lean Six Sigma Green Belt · FE / EIT (Engineer-in-Training) · Process Safety Management (PSM)

Process and CI: Lean · Six Sigma (DMAIC) · Kaizen · PDCA · OEE and loss trees · SPC · RCA · FMEA / pFMEA · Centerline management · Daily Management System (CIL / CLM / 5S) · Agile / Scrum

Data and engineering systems: Microsoft Office (Excel, Project, Power BI, Power Query) · Minitab · SQL / PostgreSQL · Grafana · Python · @RISK · AutoCAD and plant layout · PLC tag validation and historian data · SAP · Plant utility systems (chillers, glycol loops, heat exchangers, compressed air, condensate return)

Applied AI and analytics: Built an XGBoost classifier and AWS SageMaker pipeline predicting the process conditions that precede manual dosing interventions on a filling line; designs Claude-based workflows for kaizen documentation and capital one-pagers.

Languages: English (native) · Spanish (native); delivers technical training and executive presentations in both.